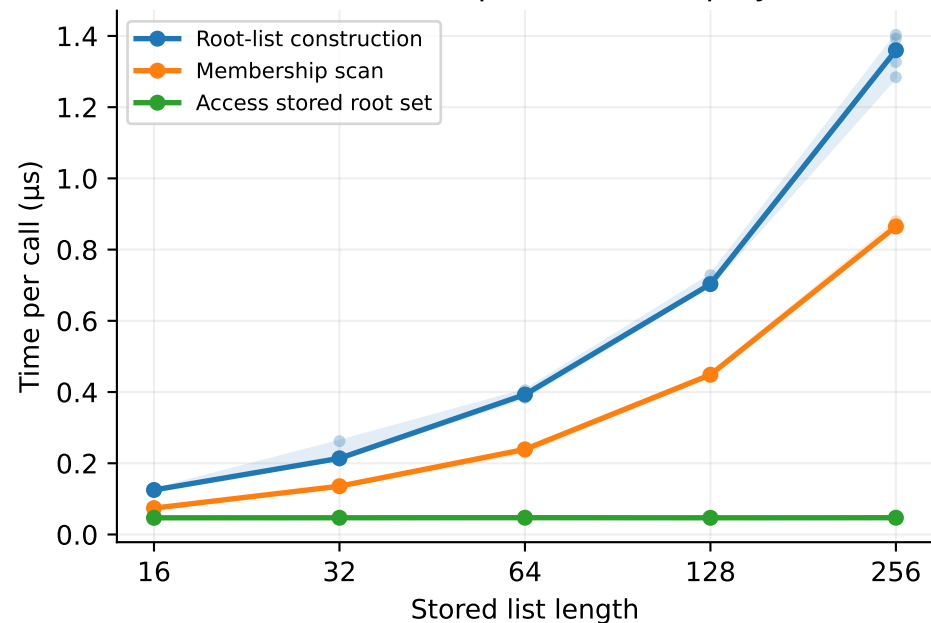
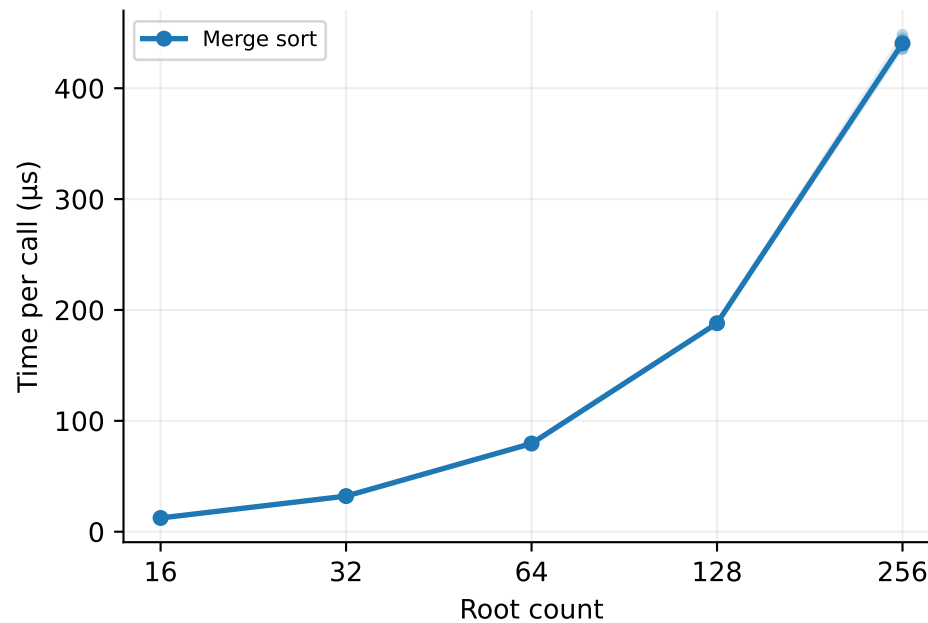


Real algebraic operations: cheap list handling, expensive canonical arithmetic

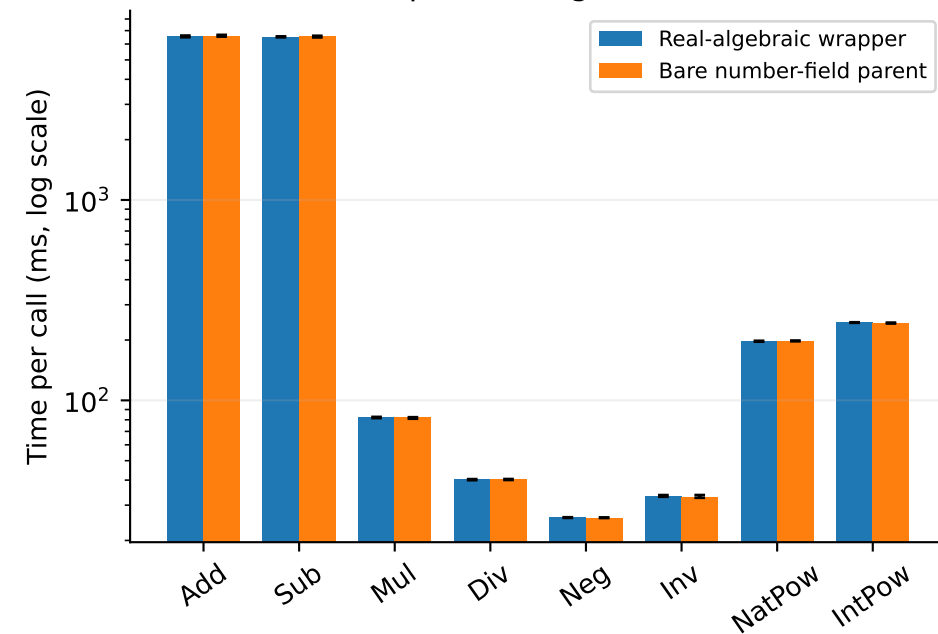
Stored roots of a repeated-factor polynomial



Sort distinct rational root records



Fixed inputs of degrees 6 and 2



Shared host; all retained trials shown, including leading points from earlier model checks. Sources: 4f44b806 (lists), 6d78bf3 (sort/arithmetic).

Arithmetic fixture: positive root of X^6-2 and $\sqrt{3}$; Add/Sub yield degree 12 and take ≈ 6.5 s. Fixed-size bars do not establish a growth rate. Paired wrappers return identical complete hashes.